UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS
General Certificate of Education Ordinary Level

BIOLOGY **5090/01**

Paper 1 Multiple Choice **October/November 2008**

1 hour

Additional Materials: Multiple Choice Answer Sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

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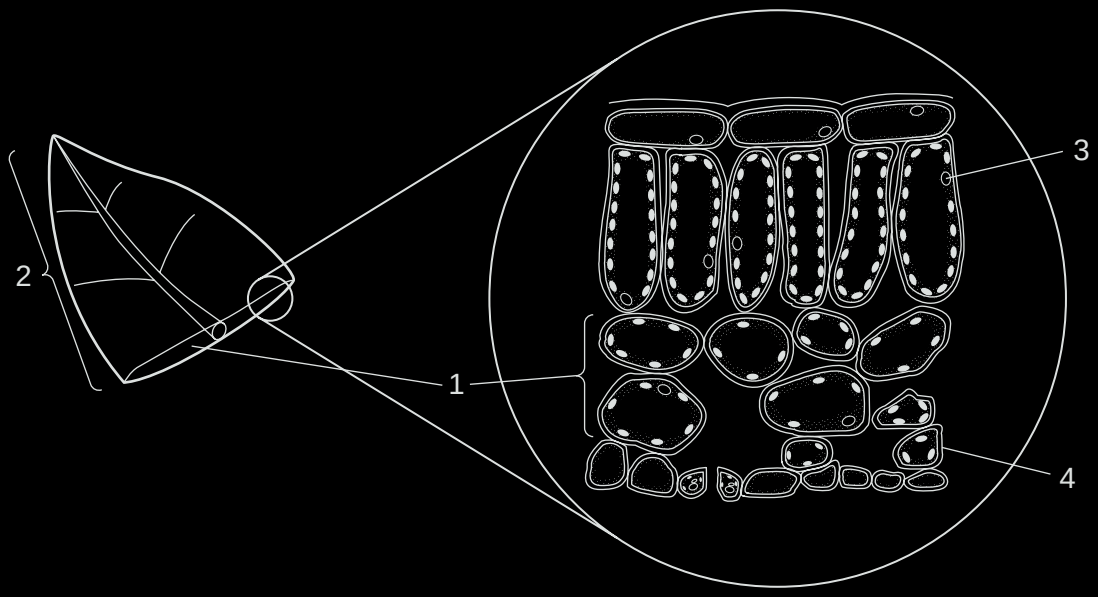
READ THESE INSTRUCTIONS FIRST

Write in soft pencil.
Do not use staples, paper clips, highlighters, glue or correction fluid.
Write your name, Centre number and candidate number on the Answer Sheet in the spaces provided unless this has been done for you.

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A**, **B**, **C** and **D**.
Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.



Which letter identifies a cell, a tissue and an organ?

	cell	tissue	organ
A	3	2	4
B	1	4	3
C	4	1	2
D	2	3	1

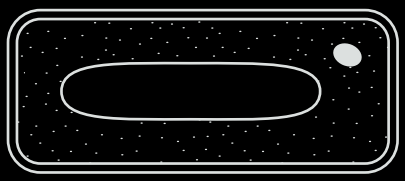


diagram 1

The cell is now placed in a concentrated sugar solution.

It changes to appear as in diagram 2.

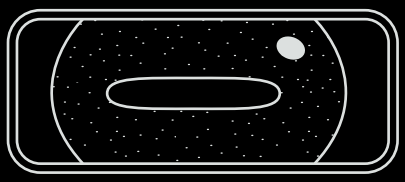


diagram 2

Which statement explains the change?

A Sugar has moved into the cell.

B Sugar has moved out of the cell.

C Water has moved into the cell.

D Water has moved out of the cell.

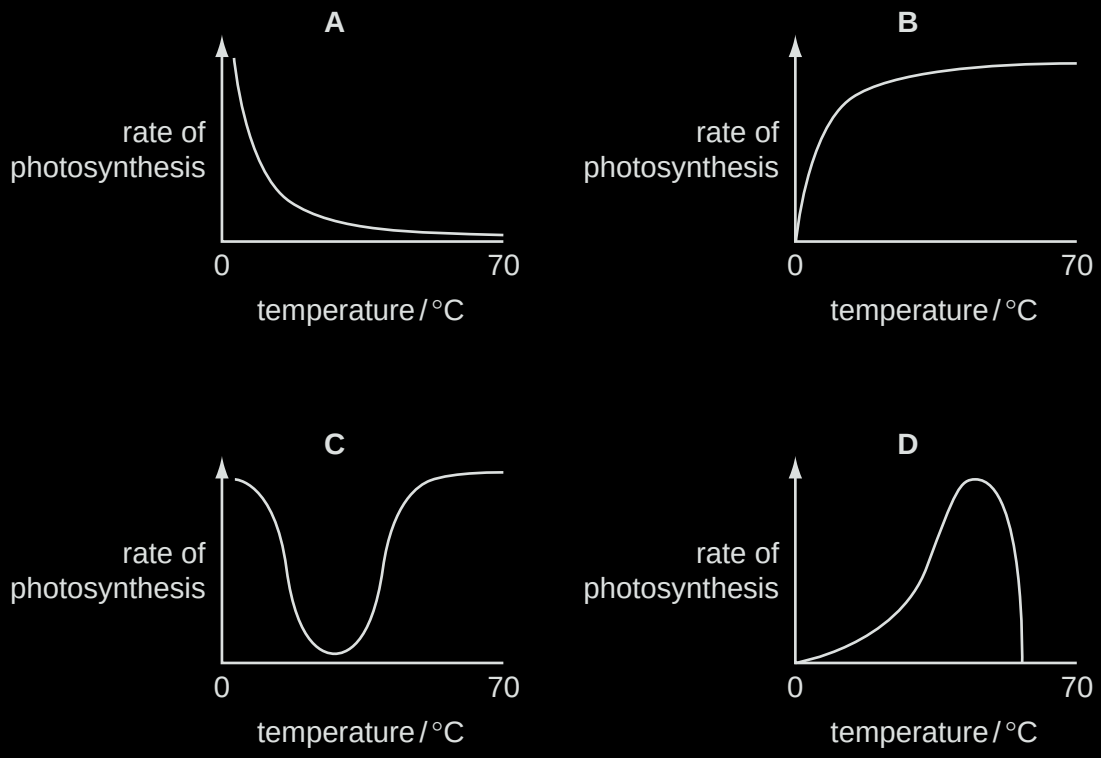
3 Four tubes containing 10 cm^3 of 1 % starch solution were treated in different ways and then mixed with saliva. After 30 minutes, 1 cm^3 of iodine in potassium iodide solution was added to each tube.

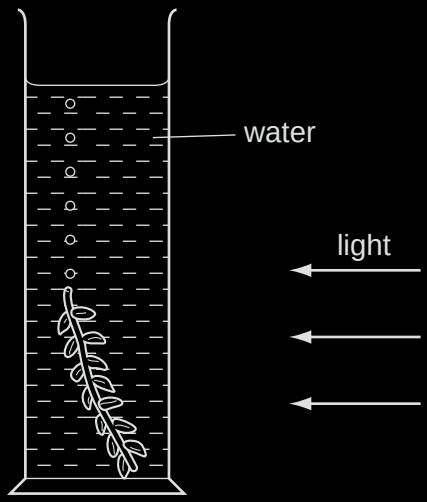
In which tubes were the contents a yellow-brown colour?

	tube incubated at 35 °C	tube incubated at 75 °C	tube incubated at pH 2.5	tube incubated at pH 6.9
A	✓		✓	
B	✓			✓
C		✓		✓
D		✓	✓	

key

✓ = yellow-brown colour


A
rate of photosynthesis
0
70
temperature / °C
B
rate of photosynthesis
0
70
temperature / °C
C
rate of photosynthesis
0
70
temperature / °C
D
rate of photosynthesis
0
70
temperature / °C



After a few minutes the bubbles cease.

Which factor in the water might be limiting the rate of photosynthesis?

A carbon dioxide

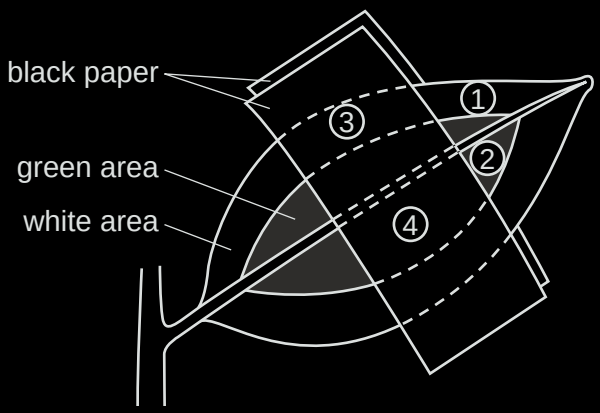
B magnesium

C nitrate

D oxygen

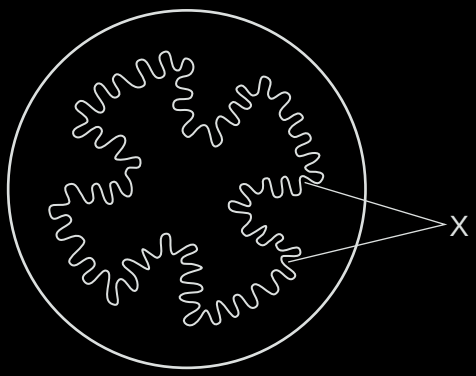
7 A plant has leaves that are partly green and partly white. The plant is destarched and a leaf is partly covered by black paper.

The plant is placed in bright light for several hours. Four discs are then cut from the leaf in the positions shown and are tested for starch.



Which discs contain starch?

A 1 only **B** 1 and 2 **C** 2 only **D** 3 and 4



What is the role of the structures labelled X?

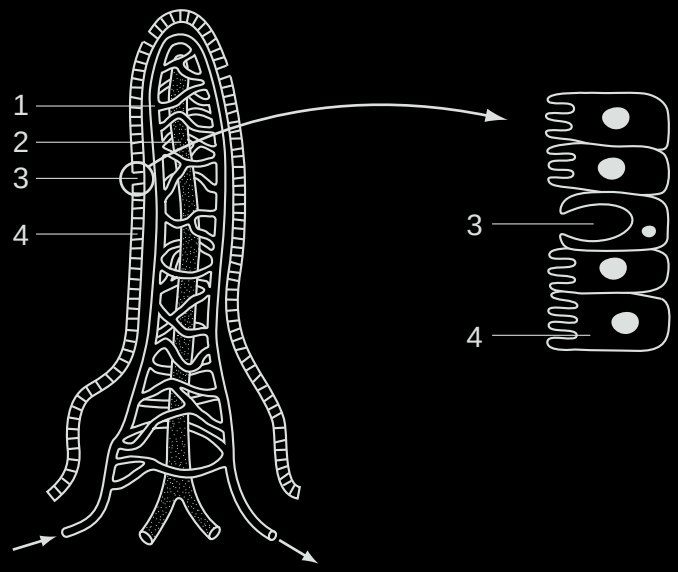
A They help to move the food along.

B They make a large surface area for absorption.

C They protect against bacteria.

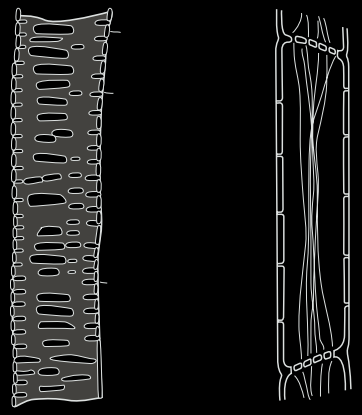
D They move mucus over the surface.

9 The diagram represents a villus.



Which sequence correctly describes the functions of the numbered parts?

	1	2	3	4
A	absorbs digested fats	absorbs glucose	produces enzymes	produces mucus
B	absorbs digested fats	absorbs glucose	produces mucus	produces enzymes
C	absorbs glucose	absorbs digested fats	produces enzymes	produces mucus
D	absorbs glucose	absorbs digested fats	produces mucus	produces enzymes



X Y

Samples of the contents of X and Y were tested.

What results are expected?

	X		Y	
	Benedict's reagent	iodine	Benedict's reagent	iodine
A	+	+	+	+
B	+	–	+	–
C	–	+	–	+
D	–	–	–	–

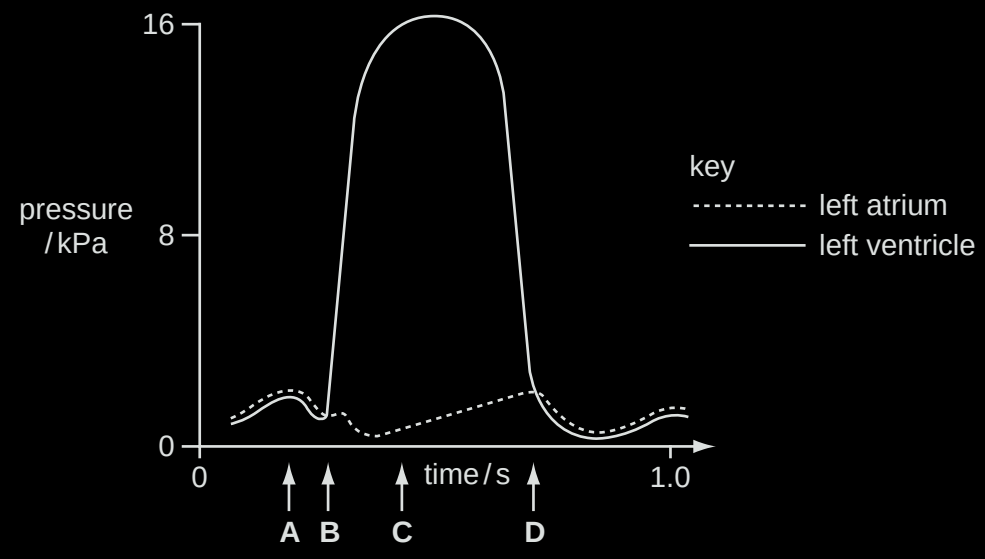
key

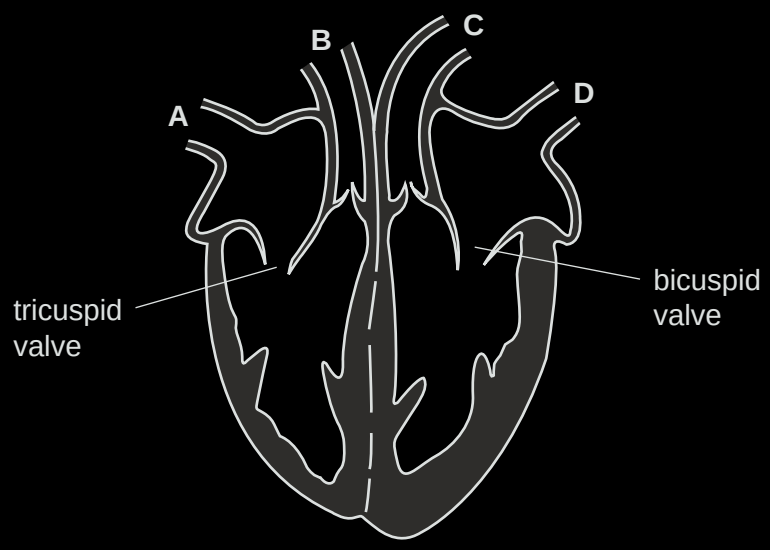
– = negative result

+ = positive result

13 The graph shows the pressure changes in the left atrium and the left ventricle while the heart is beating.

At which point do the atrio-ventricular (bicuspid and tricuspid) valves close?





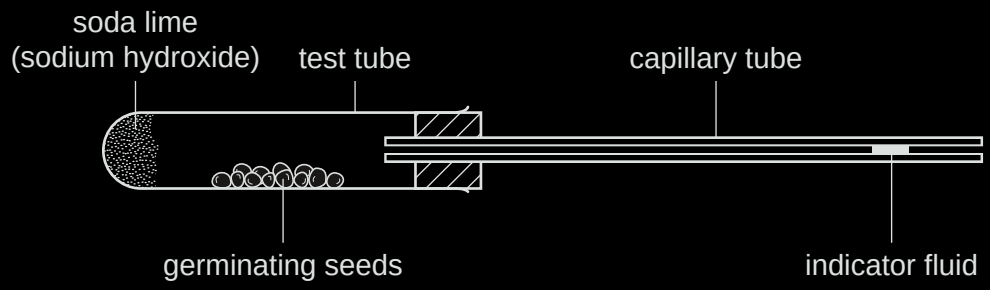
16 Which equation represents anaerobic respiration in yeast?

A glucose → alcohol + carbon dioxide

B glucose → alcohol + water

C glucose → lactic acid + carbon dioxide

D glucose → lactic acid + water



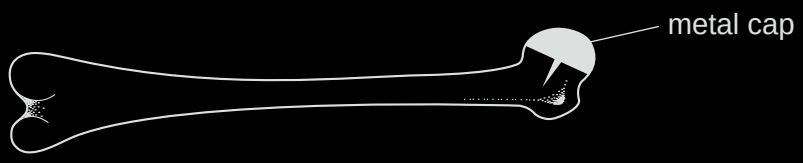
The indicator fluid in the capillary tube shown in the diagram above will

A move away from the test-tube because of oxygen output by the seeds.

B move towards the test-tube because of carbon dioxide intake by the seeds.

C move towards the test-tube because of oxygen intake by the seeds.

D remain stationary, because carbon dioxide output and oxygen intake are equal.



Which bone is this, and which joint does the metal cap repair?

	bone	joint
A	humerus	elbow
B	humerus	shoulder
C	ulna	elbow
D	ulna	shoulder

20 Urea is produced in one organ, filtered from the blood by a second organ and stored inside a third organ before being expelled from the body.

Which organs carry out these functions?

	production	filtration	storage
A	kidney	bladder	liver
B	kidney	liver	bladder
C	liver	bladder	kidney
D	liver	kidney	bladder

21 What is meant by negative feedback?

A A change away from a set point causes a change back towards the set point.

B A change away from a set point causes further change away from the set point.

C A change towards a set point causes a change away from a set point.

D Changes away from a set point are prevented.

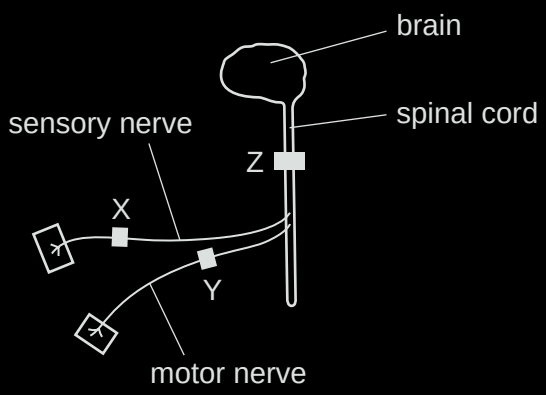
22 What is a sign of diabetes mellitus?

A glucose in the blood

B glucose in the urine

C insulin in the blood

D insulin in the urine



Of four men, one had no anaesthetic block and the other three had only one anaesthetic block at X, Y or Z.

One of the men can feel a pinprick on his leg but cannot move it.

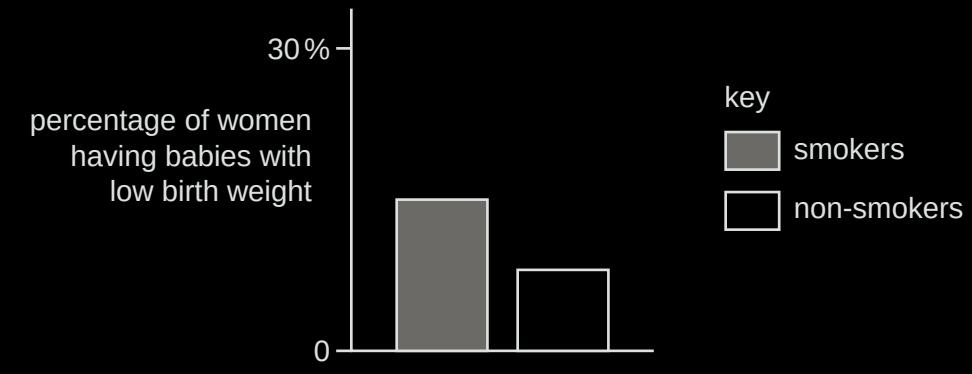
Where is the anaesthetic block?

A block is at X

B block is at Y

C block is at Z

D no block



What is shown by the bar chart?

A More women smoke during pregnancy than do not.

B Smoking during pregnancy has no effect on birth weight.

C Smoking in pregnancy increases the risk of low birth weight.

D Women whose babies have low birth weight must have smoked in pregnancy.

26 The table shows the characteristics of four microbes.

Which one could be a virus?

	contains DNA	contains one or more cells	contains one or more cell nuclei	produces spores
A	✓	✗	✗	✗
B	✓	✓	✗	✗
C	✓	✓	✓	✗
D	✓	✓	✓	✓

key
✓ = true
✗ = false

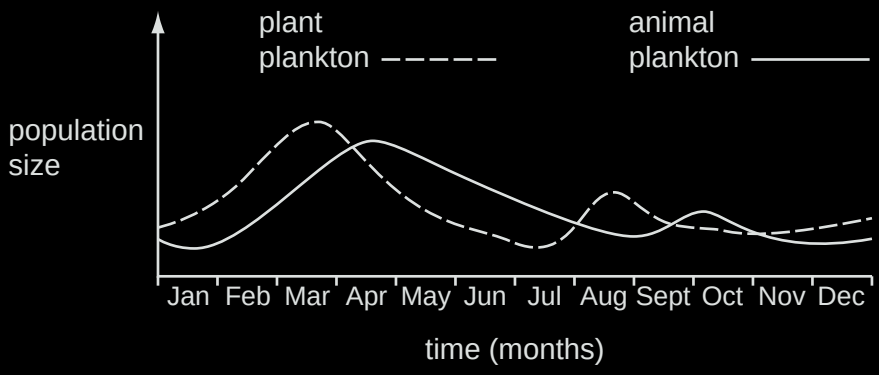
27 Which group of waste materials are likely to be decomposed most quickly by the action of micro-organisms?

A glass and building bricks

B shredded paper

C mouldy food

D toxic chemicals



Consider the following statement in relation to the data provided by the graph.

‘Population changes in animal plankton lag behind similar changes in plant plankton because the animals feed on the plants.’

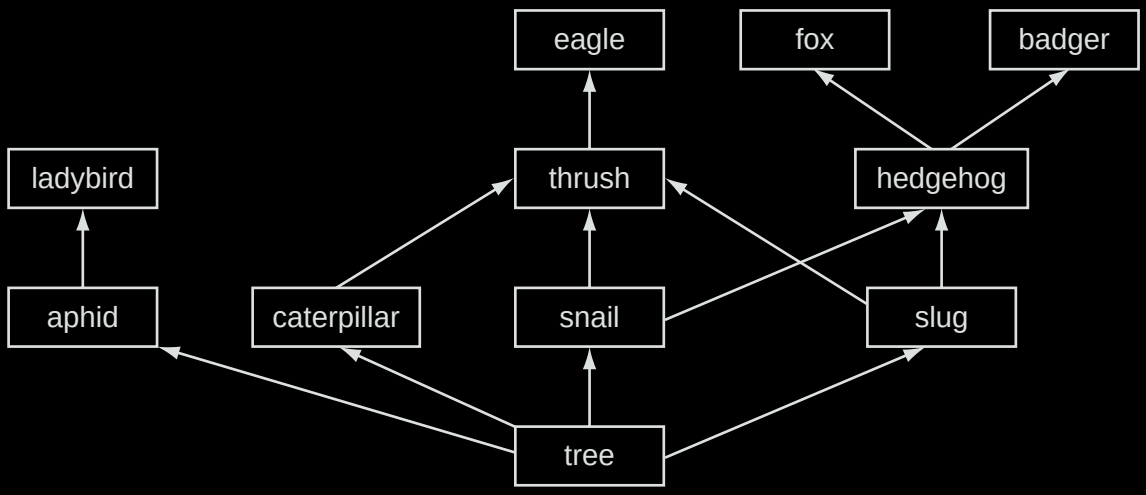
Into which category does the statement fall?

A It is a reasonable interpretation of the data.

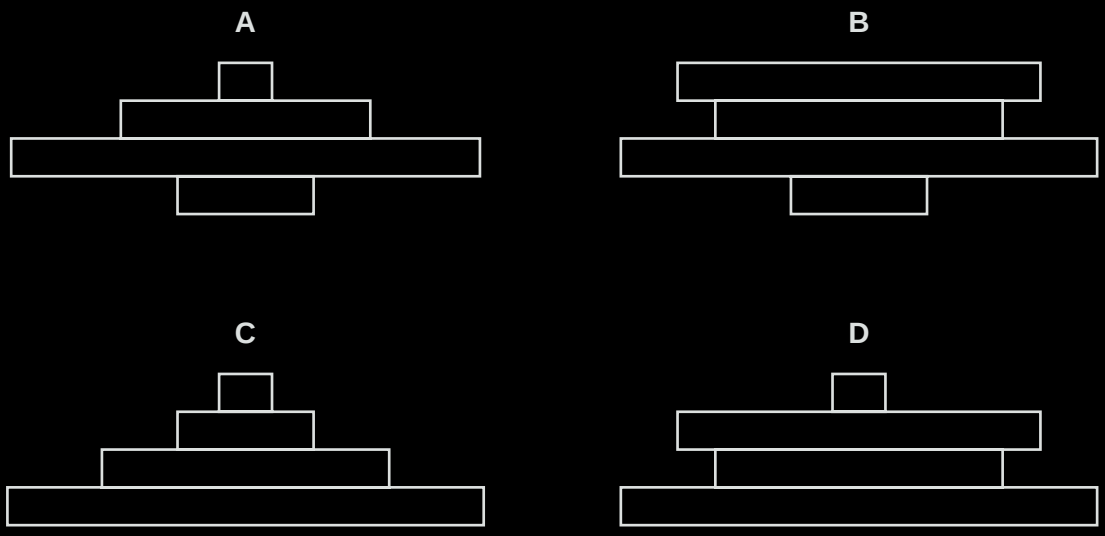
B It is a restatement of the data, not an interpretation.

C It is contradicted or not supported by the data.

D More data are required in order for this interpretation to be made.



Which is a pyramid of numbers based on this food web?



30 Which group of bacteria convert ammonia into nitrates in the nitrogen cycle?

A anaerobic bacteria

B denitrifying bacteria

C nitrifying bacteria

D nitrogen fixing bacteria

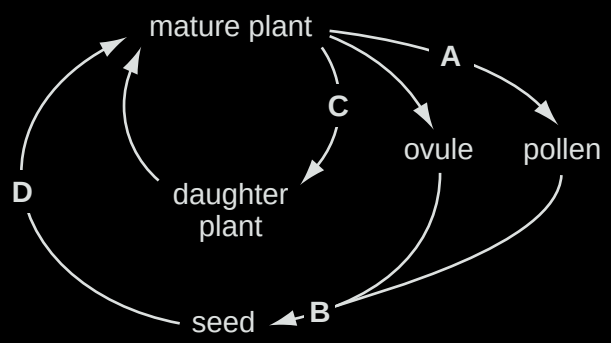
31 Which method of malarial control is effective against both adult and larval forms of the mosquito?

A covering standing water

B draining swamps

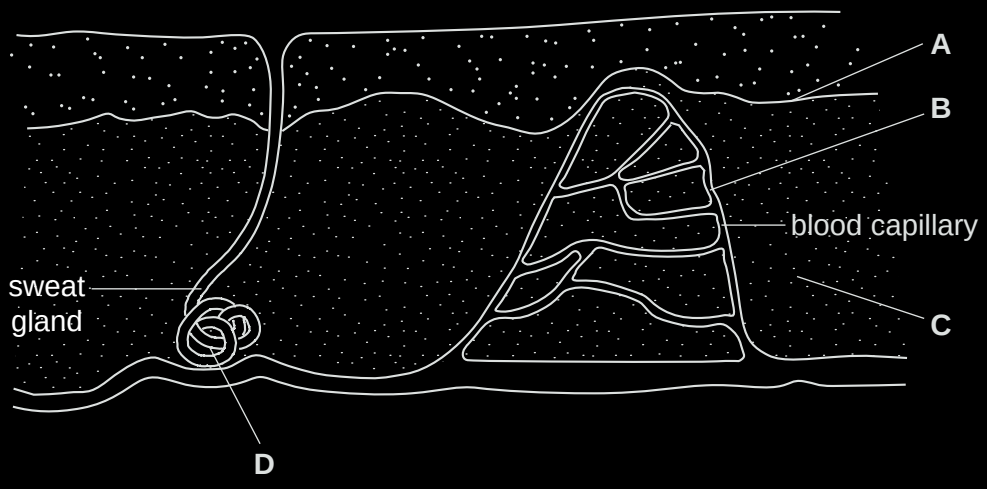
C spraying insecticides

D spraying oil on standing water



34 The diagram represents a section through the human skin.

In which part is mitosis occurring most rapidly?



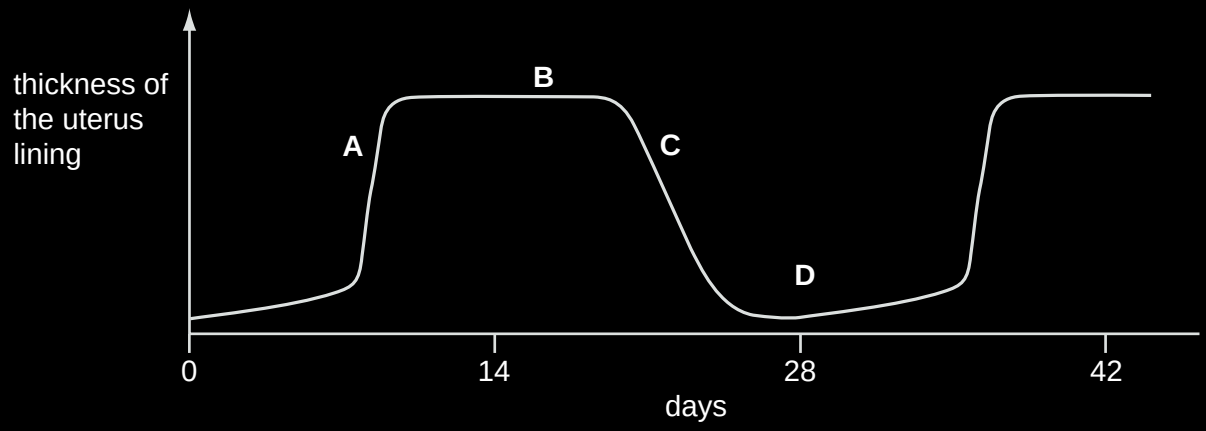
35 What is an advantage of feeding a baby with bottled milk rather than breast milk?

A Breast milk contains antibodies.

B It is easier to keep the milk free of pathogens.

C The baby's intake of food can be measured.

D The milk is always at the right temperature.



37 When a breed of cattle with red coats is crossed with the same breed with white coats, the offspring all have coats with a mixture of red and white hairs, a condition called roan.

If roan cows were crossed with a red-coated bull, the theoretical ratio of the offspring would be

A all red.

B all roan.

C 1 red : 1 roan.

D 3 red : 1 roan.

38 The diagram shows a simple genetic cross between a red flower and a white flower.

Which represents the dominant phenotype?

